

Q8

$$F_c = F_g$$

$$mr\omega^2 = \frac{GM_{blackhole}m}{r^2}$$

$$\begin{aligned}M_{blackhole} &= \frac{r^3}{G} \left(\frac{2\pi}{T}\right)^2 \\&= \frac{(1.3 \times 10^{14})^3}{6.67 \times 10^{-11}} \frac{4\pi^2}{(12 \times 365 \times 24 \times 60 \times 60)^2} \\&= 9.1 \times 10^{36} \text{ kg}\end{aligned}$$

Ans: B

Q9